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AccumulatR: Simulation and Likelihood Tools for Evidence-Accumulation Models in R

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Abstract

Evidence-accumulation models are widely used in mathematical psychology to explain response choices and response times. In practice, however, applied work often requires more than fitting a single named model family. Researchers may need architectures with multiple competing routes, pooled channels, stopping or gating processes, staged onsets, latent mixtures, or response rules that are not direct readouts of a single accumulator. **AccumulatR** is an R/C++ package designed around that modeling problem. It provides a declarative R interface for specifying accumulators, pools, triggers, timing dependencies, and observed outcome rules; it then supports simulation and likelihood evaluation for the resulting model. The user-facing workflow stays in R, while repeated numerical work is delegated to compiled C++ code. This paper gives a first software-focused overview of the package, describes its model-building vocabulary, and illustrates the basic simulation-and-fitting workflow.

Keywords: evidence accumulation, race models, response times, mathematical psychology, R.

1. Introduction: Evidence-accumulation modeling in R

Evidence-accumulation models play a central role in mathematical psychology because they provide process accounts of both response choice and response time data (Ratcliff, Smith, Brown, and McKoon 2016; Brown and Heathcote 2008; Heathcote and Matzke 2022). In many applications the practical modeling problem is not merely to estimate one standard model, but to express a psychologically motivated architecture: multiple routes can race toward the same response, some processes may be gated or inhibited, stages may begin only after earlier events, and the observed response may be defined by a logical rule rather than by the first finisher alone.

Several software tools already support important parts of this landscape. Some focus on specific model families or distributions, such as Wiener diffusion tools (Wabersich and Vandeker-

Model component	Main interface
Accumulator definition	`add_accumulator()`
Pooled channels	`add_pool()`
Observed response labels	`add_outcome()`
Logical response rules	`first_of()`, `all_of()`, `none_of()`, `inhibit()`
Triggered or gated starts	`add_trigger()`
Staged onsets	`after()`
Latent or observed mixtures	`add_component()`, `set_mixture_options()`
Multiple ranked responses	`race_spec(n_outcomes = ...)()`

Table 1: Core building blocks currently exposed by **AccumulatR**.

ckhove 2014) or recent Stan-based implementations of diffusion models (Henrich, Hartmann, Pratz, Voss, and Klauer 2024). Other packages provide highly flexible estimation frameworks, including multilevel or Bayesian modeling interfaces (Bates, Mächler, Bolker, and Walker 2015; Bürkner 2017; Vandekerckhove, Tuerlinckx, and Lee 2011). However, those interfaces do not primarily target direct specification of race-model architectures with pooled channels, trigger failures, chained onsets, and logical response rules as first-class model components.

AccumulatR was written to target that gap. The package provides a model-focused interface in R for specifying evidence-accumulation systems in terms that are close to how mathematical psychologists often describe their models: accumulators, pools, outcomes, timing relations, and mixtures. These model definitions can then be used for simulation and for likelihood evaluation against observed behavioral data. Computationally intensive repeated work is handled in C++, while the user-facing workflow remains in R (R Core Team 2025).

This draft paper is intentionally modest. Its goals are to describe the current package vocabulary, show the core workflow, and mark the sections that would need deeper development for a full *Journal of Statistical Software* submission.

2. Package overview

The basic object in **AccumulatR** is a model specification created with `race_spec()`. A specification is built incrementally and then finalized for simulation or fitting using `finalize_model()`. The resulting workflow is centered on four stages:

1. define a model architecture,
2. generate trial-wise parameters,
3. simulate or prepare behavioral data,
4. evaluate the likelihood of the observed data under candidate parameters.

At present, the package supports several accumulator distributions, including lognormal, gamma, ex-Gaussian, linear ballistic accumulator (LBA), and race diffusion model (RDM) parameterizations. In addition to accumulator-level timing parameters, the package supports several structural operators that are especially useful in mathematical psychology:

This feature set means that **AccumulatR** is not limited to a single "winner takes all" architecture. It can also represent models in which a response is produced by one of several routes, by a k-of-n pool, by the joint satisfaction of several events, or by a route that is itself vulnerable to a stop-like blocker. Likewise, mixtures can be handled either as observed trial types or as latent components marginalized in the likelihood.

3. Core workflow

The package API is deliberately compact. A minimal model is first described in terms of accumulators and outcomes, and is then turned into a finalized model object. Trial-wise parameters are generated with `build_param_matrix()`, data are either simulated with `simulate()` or supplied by the user, and likelihood evaluation then uses `prepare_data()`, `make_context()`, and `log_likelihood()`.

```
library("AccumulatR")

spec <- race_spec() |>
  add_accumulator("left", "lognormal") |>
  add_accumulator("right", "lognormal") |>
  add_outcome("left", "left") |>
  add_outcome("right", "right")

model <- finalize_model(spec)

pars <- c(
  left.m = log(0.28), left.s = 0.16,
  right.m = log(0.35), right.s = 0.18
)

param_df <- build_param_matrix(model, pars, n_trials = 500)
sim <- simulate(model, param_df, seed = 123)

prepared <- prepare_data(model, sim[c("trial", "R", "rt")])
ctx <- make_context(model)
log_likelihood(ctx, prepared, param_df)
```

This example deliberately uses a simple two-choice lognormal race so that the main workflow is clear. The same sequence of calls is used for more structured models. The package vignettes extend this pattern to pooled channels, trigger failures, chained onsets, latent mixtures, and trials with multiple observed response ranks.

One design choice worth emphasizing is the separation between structural specification and trial-wise parameters. The model object defines the architecture, while the parameter matrix defines the per-trial numerical values to be used in simulation or likelihood evaluation. This makes it straightforward to represent designs in which some parameters vary by condition or by observed trial type without changing the model architecture itself.

4. Illustrative structural extensions

The main contribution of **AccumulatR** is not the minimal race model above, but the way it generalizes beyond it. Three extensions are especially important.

First, pools allow an observed response to depend on multiple underlying accumulators. For example, a response may occur when any two of three channels have completed. This is useful when the model posits converging routes rather than a single unitary evidence stream.

Second, outcome rules can be expressed logically. A response can be generated by one of several routes, require the conjunction of several events, require the absence of another event, or be blocked by an inhibitory process. This feature is intended for models in which the observed response is a structured readout from latent events rather than a simple label attached to a single accumulator.

Third, timing dependencies can be structural rather than purely parametric. With `after()`, an accumulator can begin only when another accumulator or pool has completed. Through triggers, one can specify shared or independent failures of a gating process. These mechanisms are useful for staged-processing and stop-like architectures that are common in mathematical psychology (Heathcote, Lin, Reynolds, Strickland, Gretton, and Matzke 2019).

The package also exposes utilities that support inspection rather than only estimation. For example, `processing_tree()` provides a compact visual representation of the response logic encoded in a model, and `response_probabilities()` evaluates response probabilities implied by a model and parameter set. These tools are helpful when debugging more complex architectures.

5. Relation to existing software

At its current stage, **AccumulatR** should be viewed as complementary to existing software rather than as a replacement for it. Packages such as **RWiener** (Wabersich and Vandekerckhove 2014) and recent diffusion implementations in Stan (Henrich *et al.* 2024) provide tools that are tightly tied to particular sequential-sampling model families. More general-purpose modeling interfaces such as **lme4** (Bates *et al.* 2015) or **brms** (Bürkner 2017) offer powerful estimation environments, but they do not natively present race-model architecture itself as the central software object.

The contribution of **AccumulatR** is therefore primarily representational. It lets users describe an evidence-accumulation architecture directly, then reuse that same object for simulation, data preparation, and likelihood evaluation. For mathematical psychologists, that can be an advantage when the main scientific problem is the architecture of the process model rather than only the fitting of a standard named model.

This section is still incomplete for a full JSS submission. A stronger version should add a broader comparison to other cognitive-model software in R and beyond, including estimation-focused tools and packages aimed at hierarchical cognitive modeling (CITE).

6. Discussion

AccumulatR is an attempt to make evidence-accumulation modeling more modular on the software side. The package is built around a simple idea: model structure should be declared

explicitly in R, while repeated numerical work should be delegated to compiled code. This is useful when the scientific target is a psychologically interpretable architecture involving multiple latent channels, dependencies, and response rules.

The current draft is intentionally only a starting point. A full *Journal of Statistical Software* article should add at least three things. First, it should include a more systematic comparison to related software. Second, it should contain one or more worked examples that are closer to the kinds of tasks for which mathematical psychologists would actually use the package. Third, it should include benchmark and replication material that makes the software claims more concrete.

Even so, the present draft already captures the package's actual center of gravity: **AccumulatR** is not mainly a package for one fixed diffusion or race model, but a package for expressing and evaluating structured evidence-accumulation systems.

Acknowledgments

This is a first draft prepared from the current package state, local JSS guidance, and the package vignettes. Author affiliations, a fuller software comparison, and a worked empirical example remain to be completed.

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